

FULL STACK DEVELOPMENT – SUMMARY

CHAPTER – 5:-

- **Functions**
- **Types of Functions**
- **Return Keyword**
- **Components of a Class**
- **Types of Variables and its uses.**
- **What is Constructor**
- **Types Constructor**
- **“this” keyword**
- **Static Block & Instance Block**

FUNCTION:-

- A function is called method in java is a block of code that performs a special task.
- It helps make your code reusable,organized,and easier to read.

COMPONENTS:-

- Return type—data type of value the method returns(or void if it returns nothing).
- Method name—name of the method(follows camel case naming).
- Parameters—optional input values(like variable passed to the method).
- Method body—the code that runs when the method is call.

TYPES OF FUNCTIONS:-

- Function with no parameters and no return value.
- Function with parameters and no return value .
- Function with no parameter and a return value.
- Function with parameters and a return value.

RETURN KEYWORD:-

- ❖ If the method ‘s return type is not void you must return a value that matches the method’s declared datatype.

- ❖ If the method's return type is void, you can use return,(optional) just to exit the method early.
- ❖ Exit the method and send a value back to the caller if the method has a return type.

COMPONENTS OF A CLASS:-

- ❖ A Class in java is a blueprint that defines how an object behaves and values and what data it holds.
- ❖ Variable(data members)-store data.
- ❖ Methods(member functions)-define actions.
- ❖ Constructor—initialize objects.
- ❖ Blocks(static/instance)-execute specific code.
- ❖ Nested class- inner class(optional).

TYPES OF VARIABLES AND USES:-

- ❖ Instance variable = employee name(different for each employee).
- ❖ Static variable= company name(same for all).
- ❖ Local variable = temporary data like bonus calculations.

WHAT IS CONSTRUCTOR:-

- ❖ A Constructor is a special method used to create (initialize objects in java).
- ❖ Its name is always the same as the class.
- ❖ It does not have a return type(not even void).
- ❖ It is automatically called when you create an object using new.

TYPES OF CONSTRUCTOR:-

- ❖ Default constructor-no parameters,assigns default values.
- ❖ Parameterized constructor-takes parameters to assign default values.
- ❖ Copy constructor-copies value from another object.

THIS KEYWORD:-

- ❖ A Method needs access to the entire object, not just one variable.
- ❖ We want to send the current object to another class or method.
- ❖ We want object to object interaction.
- ❖ This refers to the current object of the class.
- ❖ It is used when local variables and instance variables have the same name.

STATIC BLOCK :-

- ❖ A Static block is executed once when the class is loaded before the main() method.
- ❖ It is mainly used to initialize static variables.

INSTANCE BLOCK:-

- ❖ An instance block is executed every time an object is created, before the constructor.
- ❖ Runs every time an object is created
- ❖ Declared as {} without static.